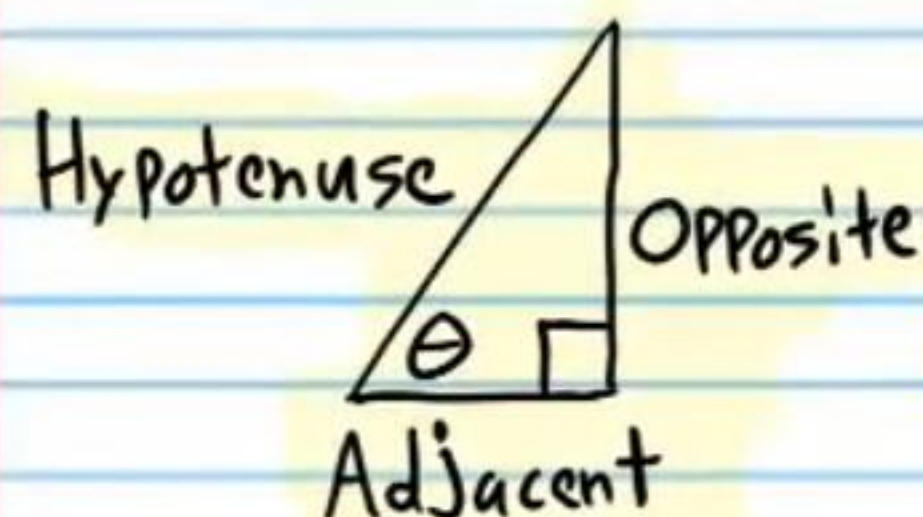


Area (Part II): Segments of Circles

Name
Date
Course

Trigonometry Review

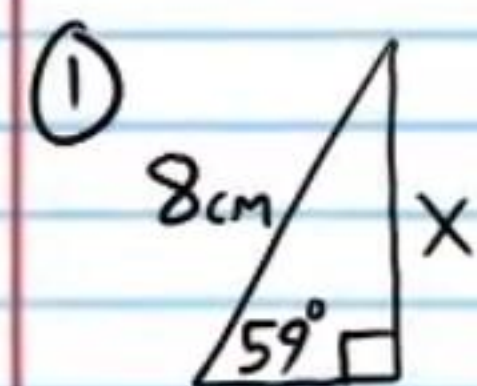


$$\sin \theta = \frac{\text{OPP}}{\text{HYP}}$$

$$\cos \theta = \frac{\text{Adj}}{\text{HYP}}$$

$$\tan \theta = \frac{\text{OPP}}{\text{Adj}}$$

Find the value of each variable below. Round to the nearest hundredth.

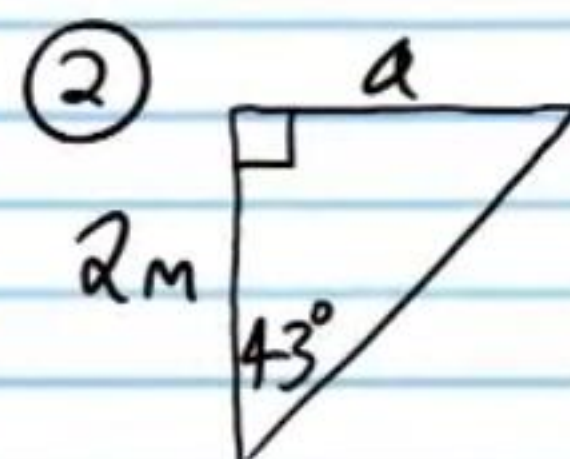


$$\sin 59^\circ = \frac{x}{8}$$

$$(8) \sin 59^\circ = \frac{x}{8} (8)$$

$$8 \sin 59^\circ = x$$

$$\boxed{6.86 \text{ cm} \approx x}$$

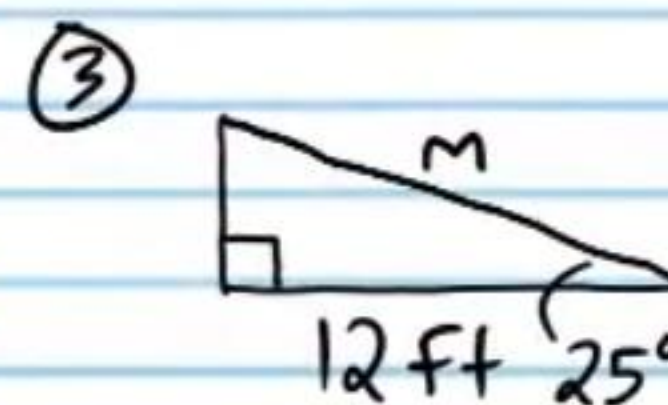


$$\tan 43^\circ = \frac{a}{2}$$

$$(2) \tan 43^\circ = \frac{a}{2} (2)$$

$$2 \tan 43^\circ = a$$

$$\boxed{1.87 \text{ m} \approx a}$$



$$\cos 25^\circ = \frac{12}{m}$$

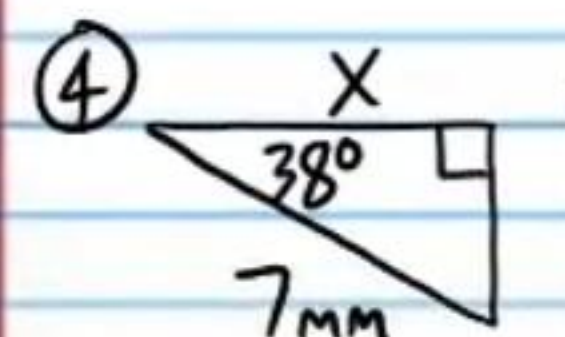
$$(m) \cos 25^\circ = \frac{12}{m} (m)$$

$$m \cos 25^\circ = 12$$

$$\frac{m \cos 25^\circ}{\cos 25^\circ} = \frac{12}{\cos 25^\circ}$$

$$m = \frac{12}{\cos 25^\circ}$$

$$\boxed{m \approx 13.24 \text{ ft}}$$

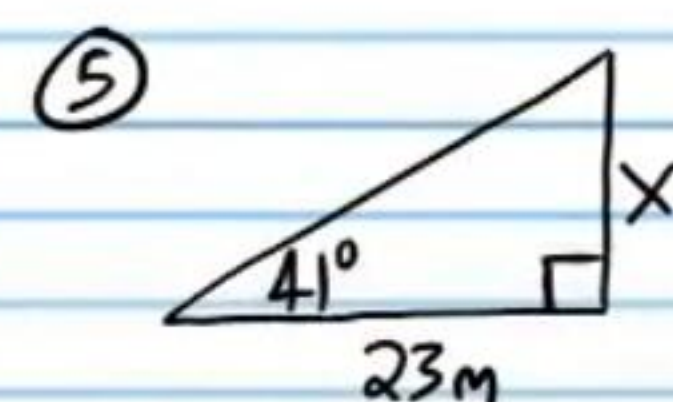


$$\cos 38^\circ = \frac{x}{7}$$

$$(7) \cos 38^\circ = \frac{x}{7} (7)$$

$$7 \cos 38^\circ = x$$

$$\boxed{5.52 \text{ mm} \approx x}$$

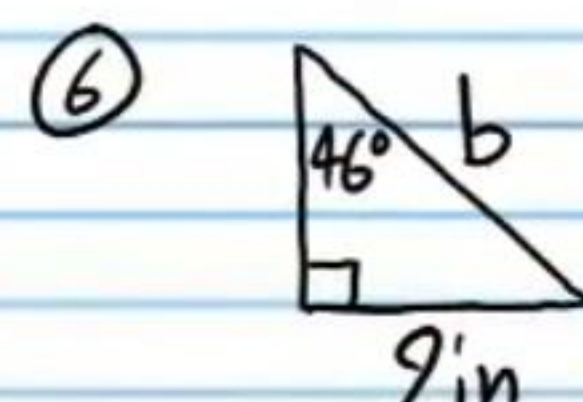


$$\tan 41^\circ = \frac{x}{23}$$

$$23 \tan 41^\circ = \frac{x}{23} (23)$$

$$23 \tan 41^\circ = x$$

$$\boxed{19.99 \text{ m} \approx x}$$



$$\sin 46^\circ = \frac{9}{b}$$

$$(b) \sin 46^\circ = \frac{9}{b} (b)$$

$$b \sin 46^\circ = 9$$

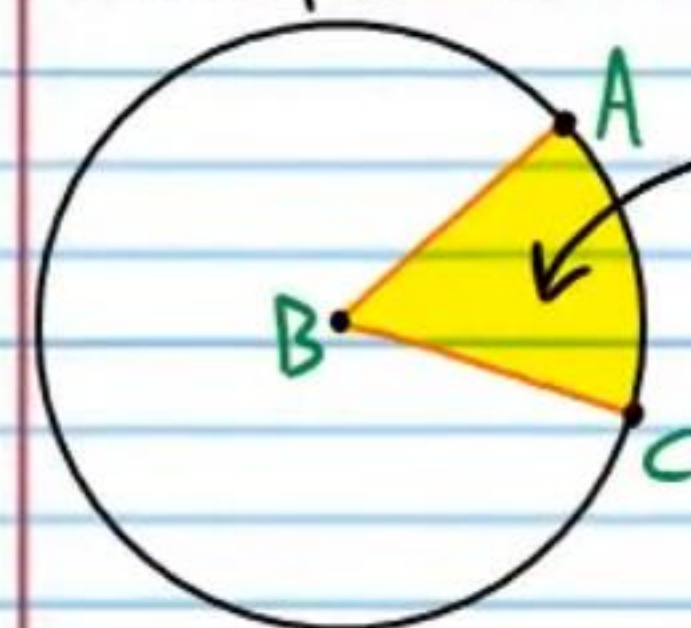
$$\frac{b \sin 46^\circ}{\sin 46^\circ} = \frac{9}{\sin 46^\circ}$$

$$b = \frac{9}{\sin 46^\circ}$$

$$\boxed{b \approx 12.51 \text{ in}}$$

Sector Review

Example: Sector ABC below.



Sector: A region in a circle bounded by two radii and the corresponding arc.

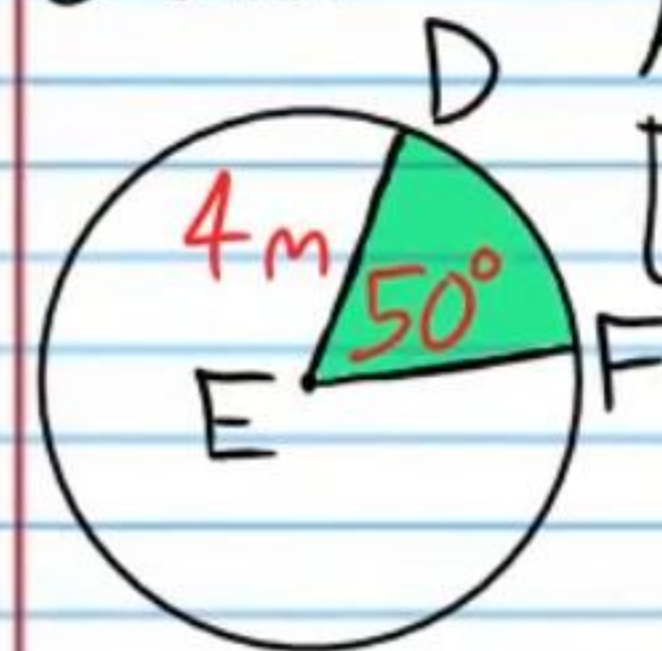
Area of a Sector

$$A = \frac{\theta}{360} (\pi r^2)$$

θ is central angle and r is radius

If E is the center of the following circles, find the area of the sectors below. You must use the π button on your calculator. Round to the nearest hundredth.

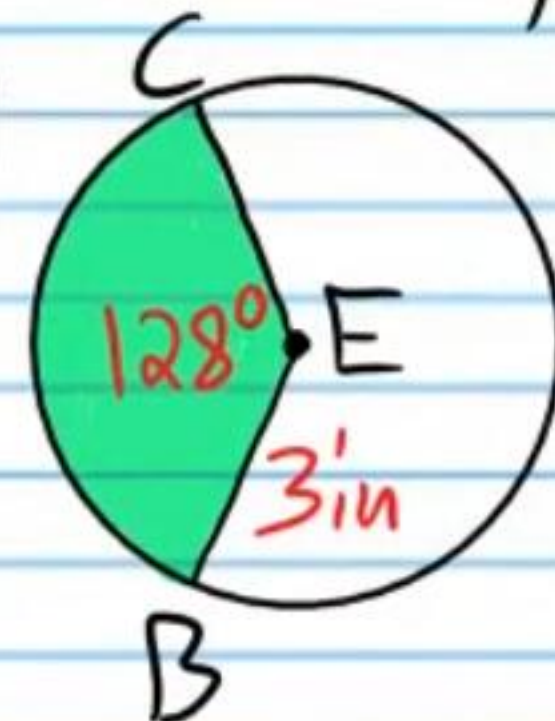
⑦ DEF



$$A = \frac{50}{360} \pi (4)^2$$

$$A \approx 6.98 \text{ m}^2$$

⑧ CEB

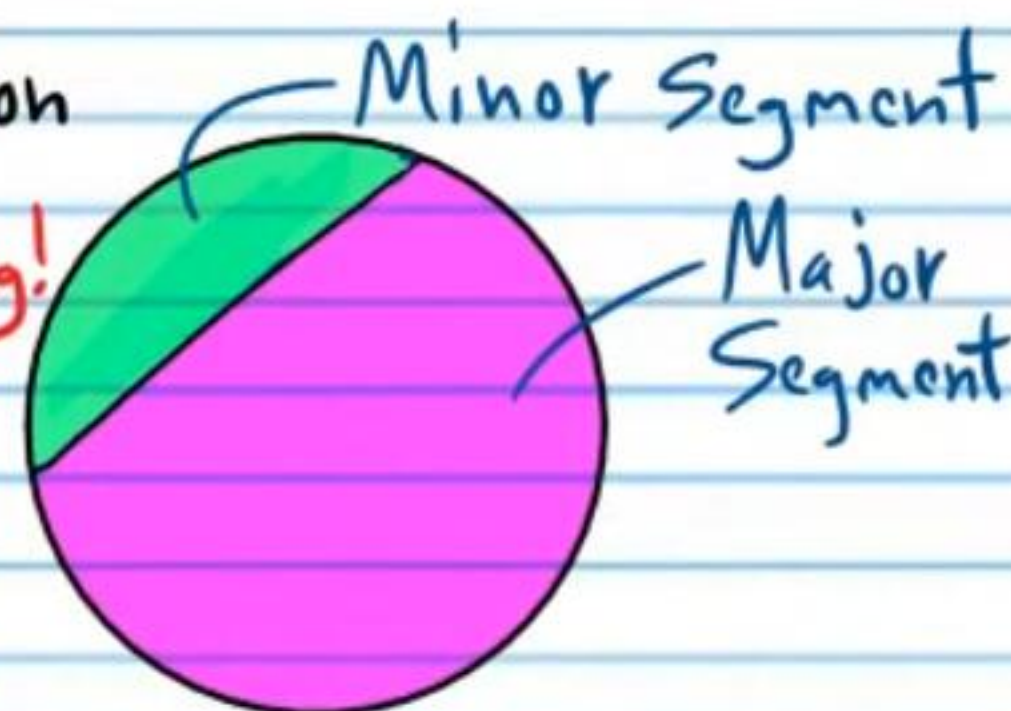


$$A = \frac{128}{360} \pi (3)^2$$

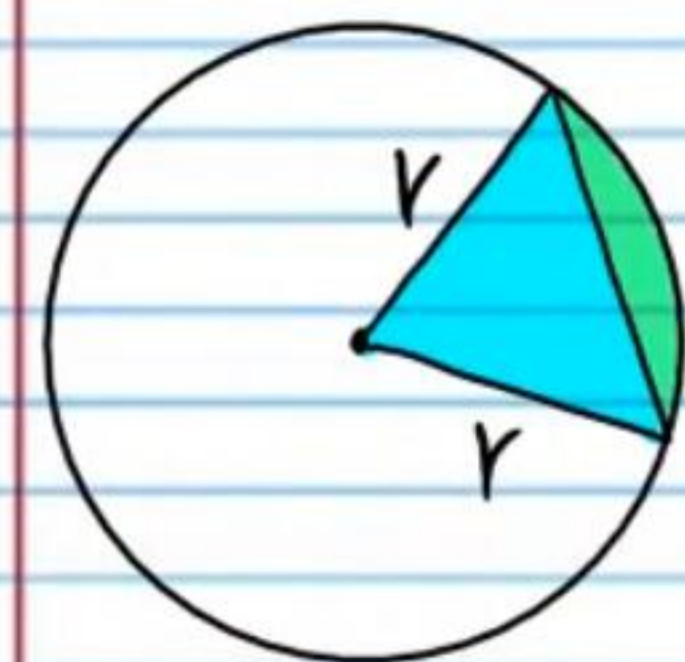
$$A \approx 10.05 \text{ in}^2$$

Segment of a Circle

Segment of a Circle: A region bounded by a circle and a chord of the circle. **Warning!** "Segment" is the same word that is used for a one dimensional line segment.

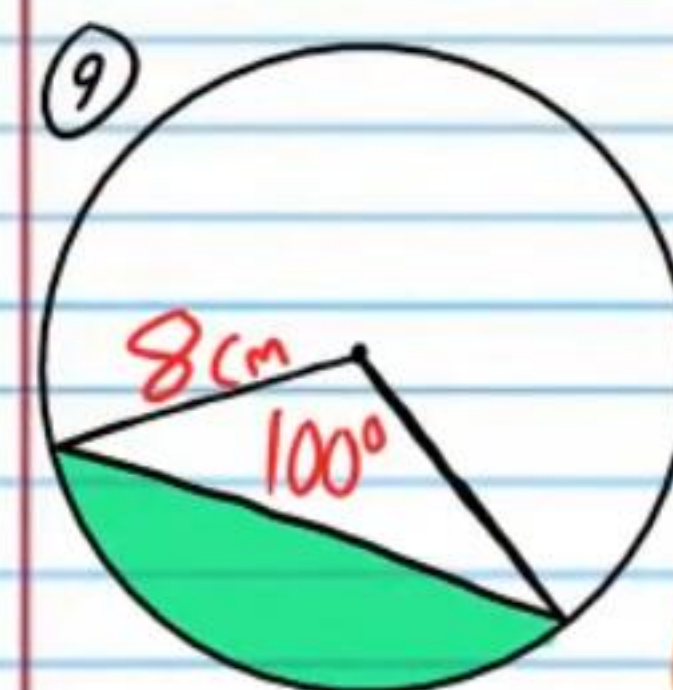


Area of a Segment of a Circle



$$A_{\text{segment}} = A_{\text{sector}} - A_{\text{triangle}}$$

Find the area of each segment below. Round to the nearest hundredth.



$$\text{I) } A_{\text{sector}} = \frac{100}{360} \pi (8)^2 \approx 55.8505$$

$$\text{II) } A_{\text{triangle}} = \frac{1}{2}bh$$

$$\cos 50^\circ = \frac{h}{8}$$

$$8 \cos 50^\circ = h$$

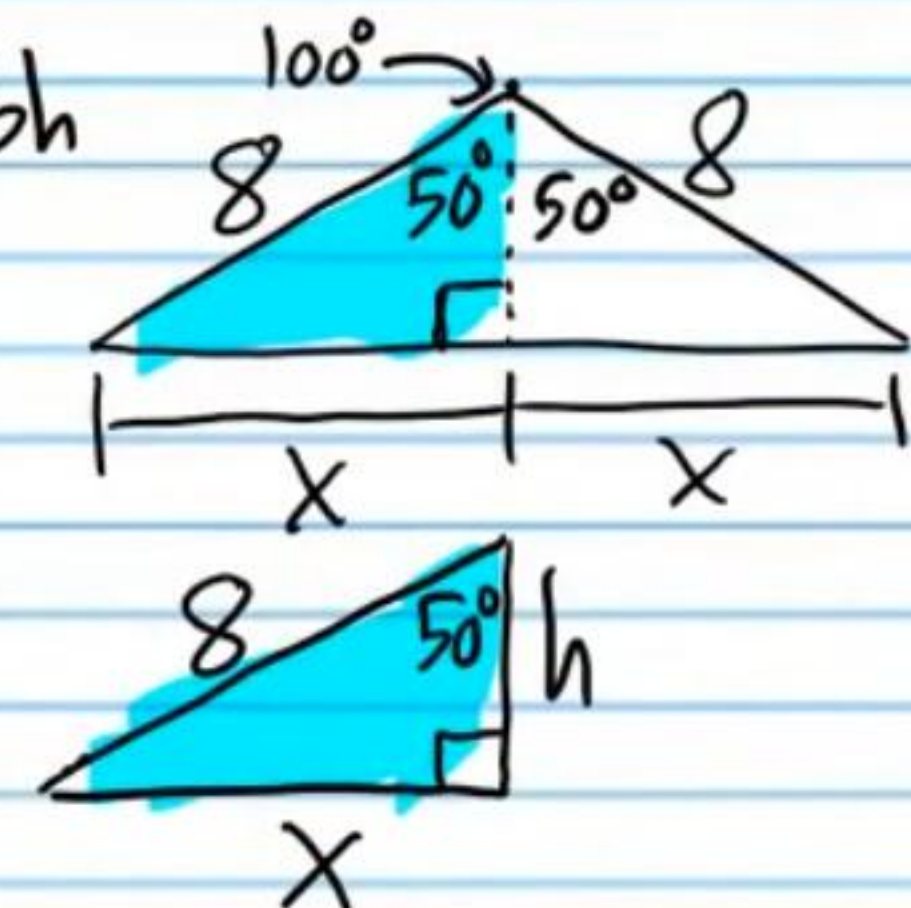
$$\frac{\text{Base}}{\sin 50^\circ} = \frac{x}{8}$$

$$8 \sin 50^\circ = x$$

$$b = 2x$$

$$b = 2(8 \sin 50^\circ)$$

$$b = 16 \sin 50^\circ$$



$$\rightarrow A_{\text{tri}} = \frac{1}{2}(16 \sin 50^\circ)(8 \cos 50^\circ)$$

$$A_{\text{tri}} \approx 31.5138$$

$$\text{III) } A_{\text{seg}} = A_{\text{sector}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 55.8505 - 31.5138$$

$$A_{\text{seg}} \approx 24.34 \text{ cm}^2$$

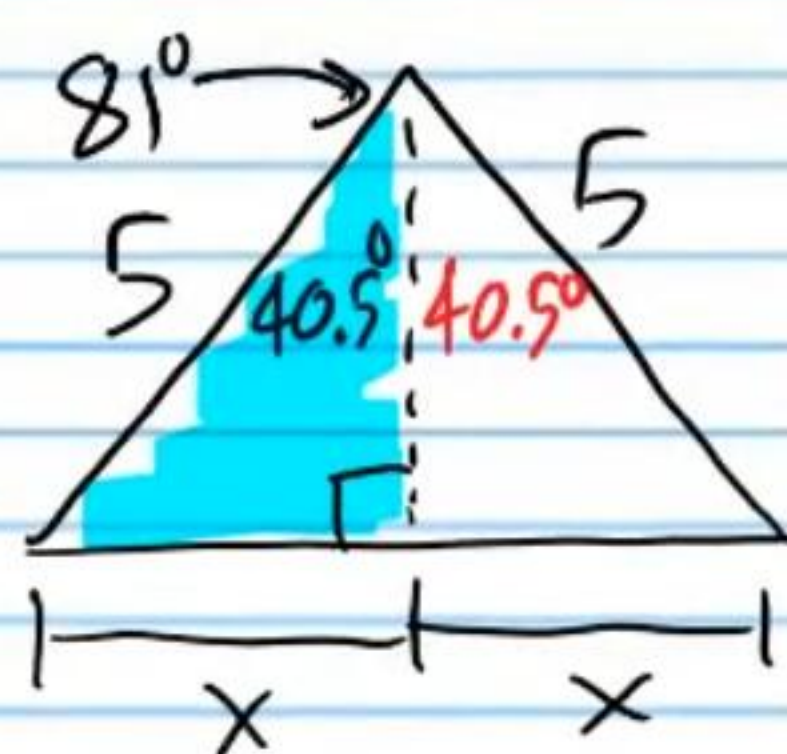
⑩ I) $A_{\text{sector}} = \frac{81}{360} \pi (5)^2 \approx 17.6714$



II) $A_{\text{triangle}} = \frac{1}{2}bh$

Height
 $\cos 40.5^\circ = \frac{h}{5}$

$5 \cos 40.5^\circ = h$



Base

$\sin 40.5^\circ = \frac{x}{5}$

$5 \sin 40.5^\circ = x$

$b = 2x$

$b = 2(5 \sin 40.5^\circ)$

$b = 10 \sin 40.5^\circ$

$A_{\text{tri}} = \frac{1}{2}(10 \sin 40.5^\circ)(5 \cos 40.5^\circ)$

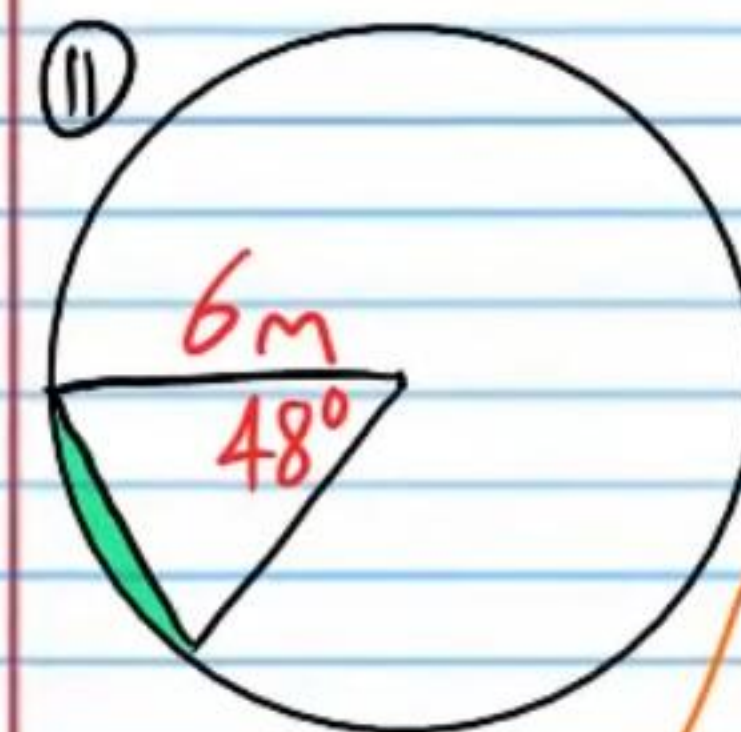
$A_{\text{tri}} \approx 12.3461$

III) $A_{\text{seg}} = A_{\text{sect}} - A_{\text{tri}}$

$A_{\text{seg}} \approx 17.6714 - 12.3461$

$A_{\text{seg}} \approx 5.33 \text{ ft}^2$

⑪ I) $A_{\text{sector}} = \frac{48}{360} \pi (6)^2 \approx 15.0796$



II) $A_{\text{triangle}} = \frac{1}{2}bh$

Height
 $\cos 24^\circ = \frac{h}{6}$

$6 \cos 24^\circ = h$



Base

$\sin 24^\circ = \frac{x}{6}$

$6 \sin 24^\circ = x$

$b = 2x$

$b = 2(6 \sin 24^\circ)$

$b = 12 \sin 24^\circ$

$A_{\text{tri}} = \frac{1}{2}(12 \sin 24^\circ)(6 \cos 24^\circ)$

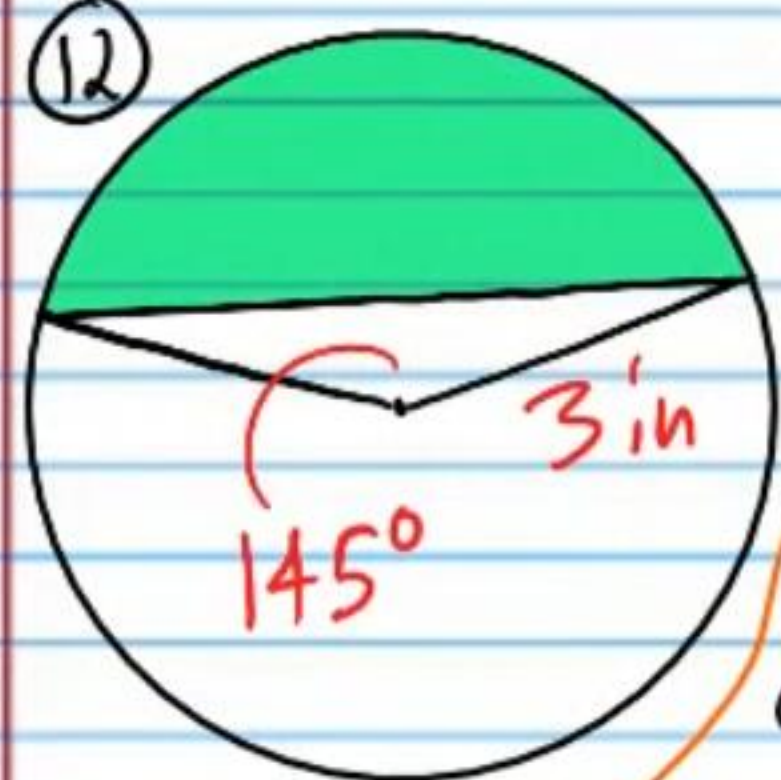
$A_{\text{tri}} \approx 13.3766$

III) $A_{\text{seg}} = A_{\text{sect}} - A_{\text{tri}}$

$A_{\text{seg}} \approx 15.0796 - 13.3766$

$A_{\text{seg}} \approx 1.7 \text{ m}^2$

⑫



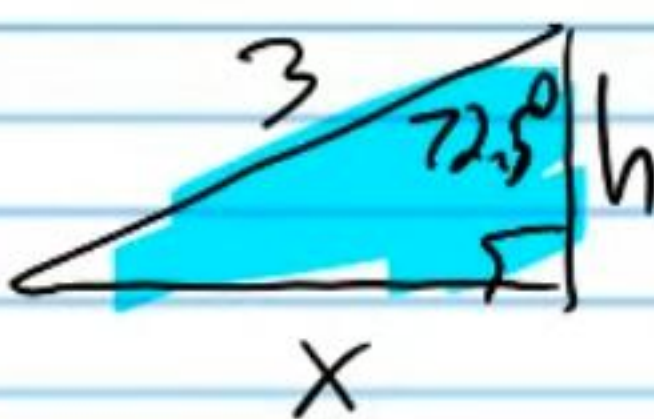
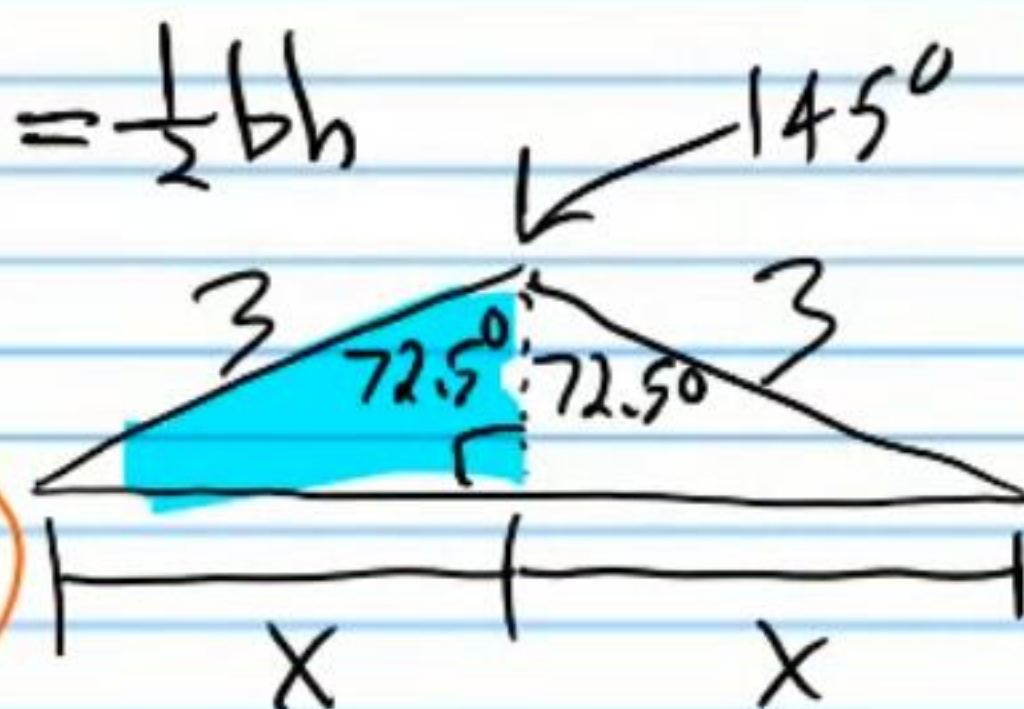
$$I) A_{\text{sector}} = \frac{145}{360} \pi (3)^2 \approx 11.3882$$

$$II) A_{\text{triangle}} = \frac{1}{2}bh$$

Height

$$\cos 72.5^\circ = \frac{h}{3}$$

$$3 \cos 72.5^\circ = h$$



Base

$$\sin 72.5^\circ = \frac{x}{3}$$

$$3 \sin 72.5^\circ = x$$

$$b = 2x$$

$$b = 2(3 \sin 72.5^\circ)$$

$$b = 6 \sin 72.5^\circ$$

$$A_{\text{tri}} = \frac{1}{2}(6 \sin 72.5^\circ)(3 \cos 72.5^\circ)$$

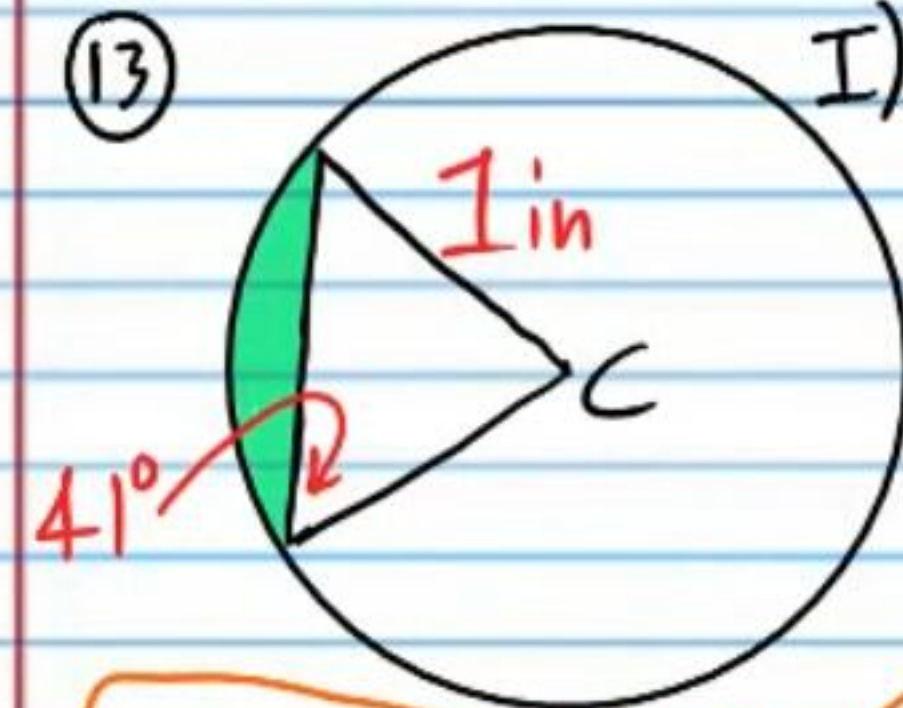
$$A_{\text{tri}} \approx 2.581$$

$$III) A_{\text{seg}} = A_{\text{sect}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 11.3882 - 2.581$$

$$A_{\text{seg}} \approx 8.81 \text{ in}^2$$

⑬



$$I) m\angle C + 41 + 41 = 180$$

$$m\angle C = 98^\circ$$

$$II) A_{\text{sector}} = \frac{98}{360} \pi (1)^2 \approx 0.8552$$

$$III) A_{\text{triangle}} = \frac{1}{2}bh$$

Height

$$\cos 49^\circ = \frac{h}{1}$$

$$\cos 49^\circ = h$$

Base

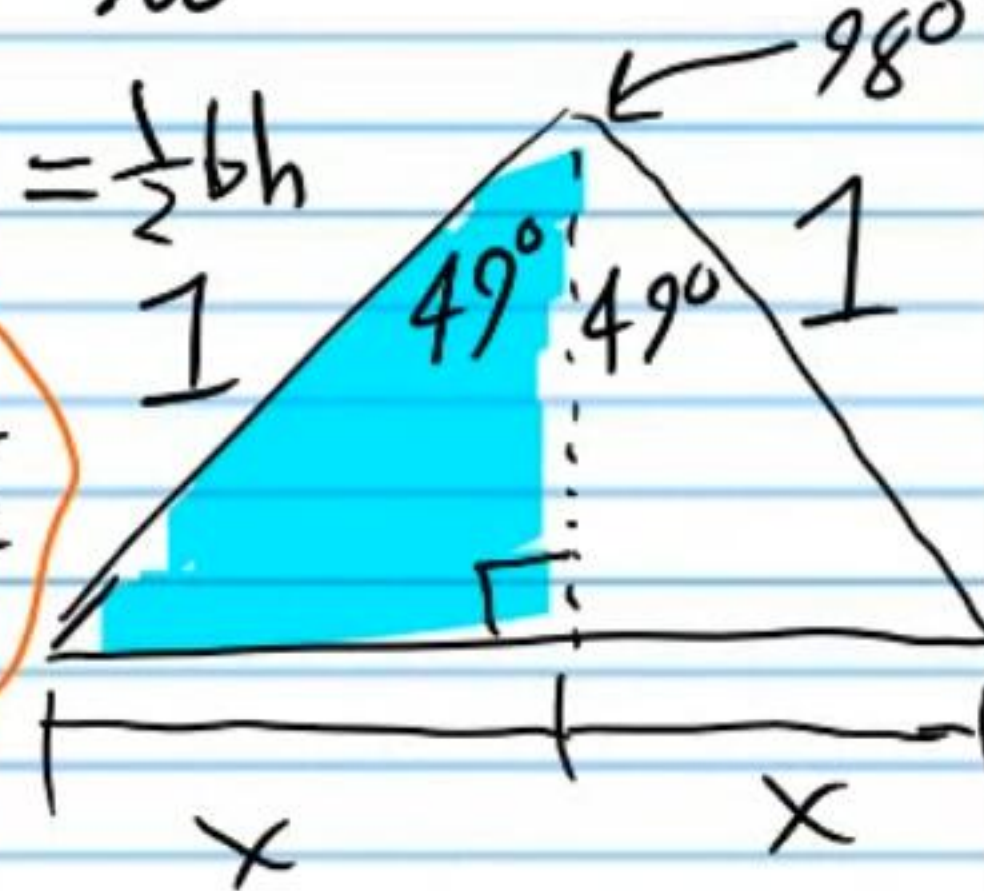
$$\sin 49^\circ = \frac{x}{1}$$

$$\sin 49^\circ = x$$

$$b = 2x$$

$$b = 2(\sin 49^\circ)$$

$$b = 2 \sin 49^\circ$$



$$A_{\text{tri}} = \frac{1}{2}(2 \sin 49^\circ)(\cos 49^\circ)$$

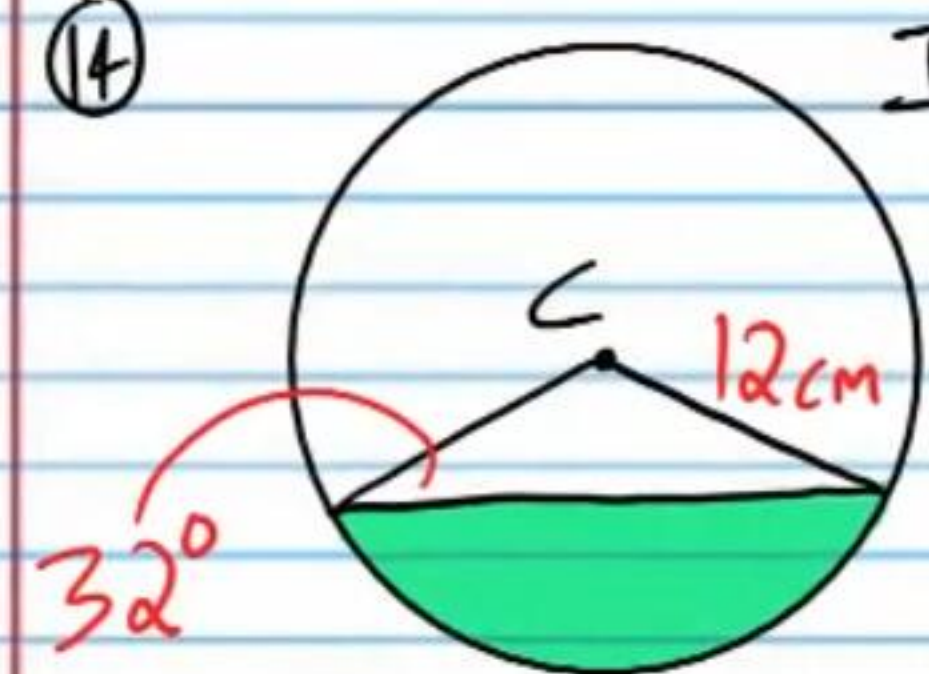
$$A_{\text{tri}} \approx 0.4951$$

$$IV) A_{\text{seg}} = A_{\text{sect}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 0.8552 - 0.4951$$

$$A_{\text{seg}} \approx 0.36 \text{ in}^2$$

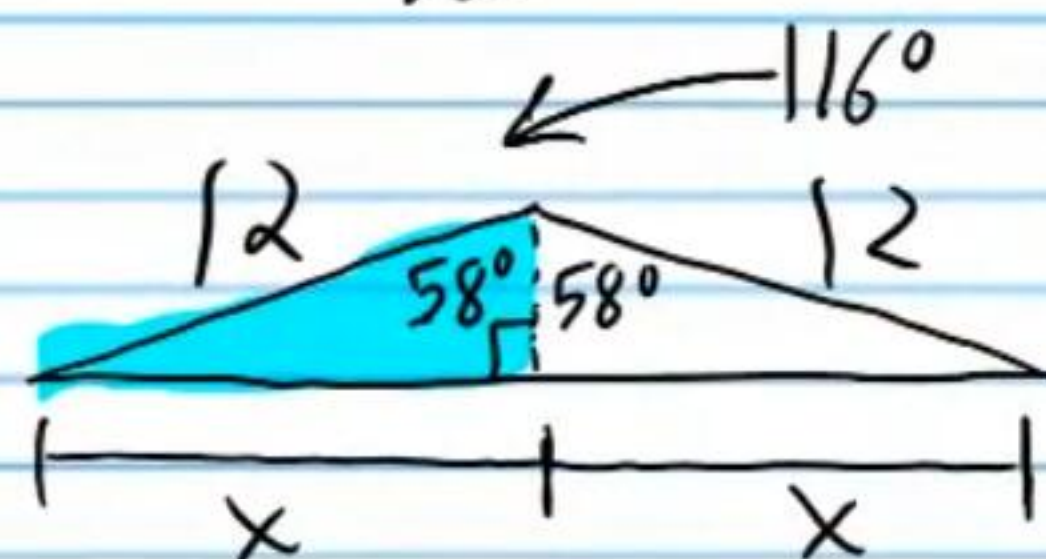
⑭



$$I) m\angle C + 32 + 32 = 180$$

$$m\angle C = 116^\circ$$

$$II) A_{\text{sector}} = \frac{116}{360} \pi (12)^2 \approx 145.7698$$



$$III) A_{\text{triangle}} = \frac{1}{2}bh$$

Height

$$\cos 58^\circ = \frac{h}{12}$$

$$12 \cos 58^\circ = h$$

Base

$$\sin 58^\circ = \frac{x}{12}$$

$$12 \sin 58^\circ = x$$

$$b = 2x$$

$$b = 2(12 \sin 58^\circ)$$

$$b = 24 \sin 58^\circ$$

$$A_{\text{tri}} = \frac{1}{2}(24 \sin 58^\circ)(12 \cos 58^\circ)$$

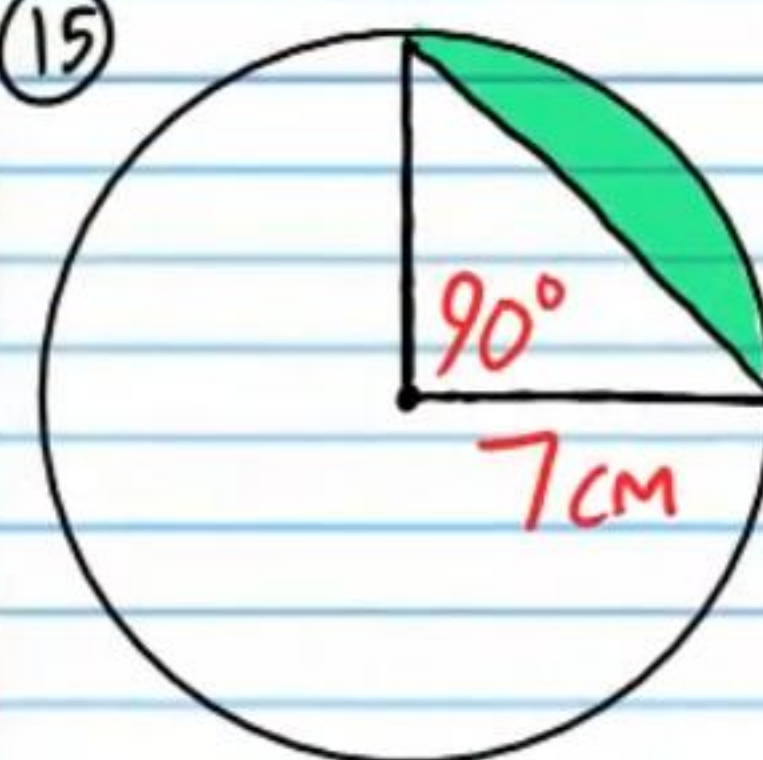
$$A_{\text{tri}} \approx 64.7131$$

$$IV) A_{\text{seg}} = A_{\text{sector}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 145.7698 - 64.7131$$

$$A_{\text{seg}} \approx 81.06 \text{ cm}^2$$

⑮



$$I) A_{\text{sector}} = \frac{90}{360} \pi (7)^2 \approx 38.4845$$

$$II) A_{\text{triangle}} = \frac{1}{2}bh$$

$$A_{\text{tri}} = \frac{1}{2}(7)(7)$$

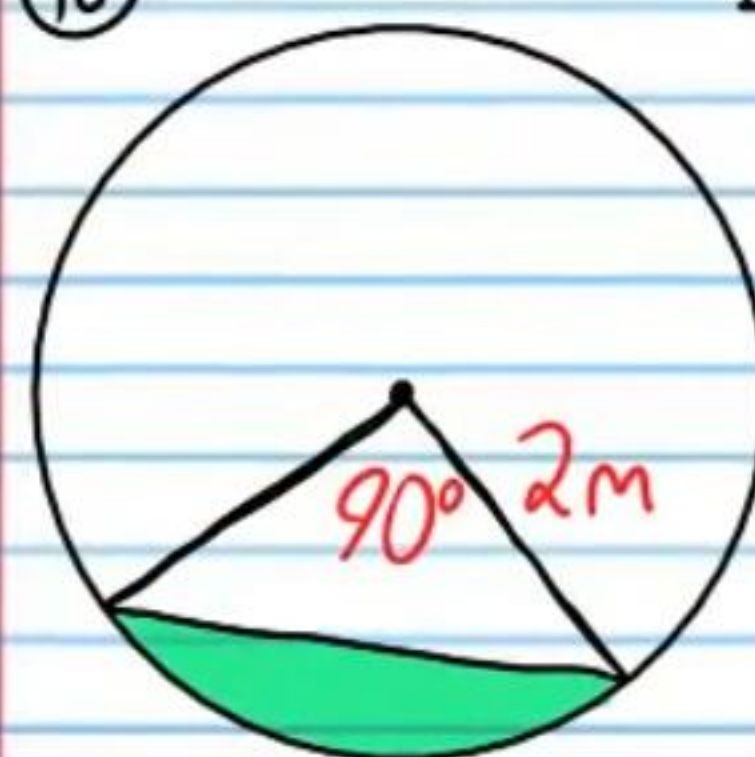
$$A_{\text{tri}} = 24.5$$

$$III) A_{\text{seg}} = A_{\text{sector}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 38.4845 - 24.5$$

$$A_{\text{seg}} \approx 13.98 \text{ cm}^2$$

⑯



$$I) A_{\text{sector}} = \frac{90}{360} \pi (2)^2 \approx 3.1415$$

$$II) A_{\text{triangle}} = \frac{1}{2}bh$$

$$A_{\text{tri}} = \frac{1}{2}(2)(2)$$

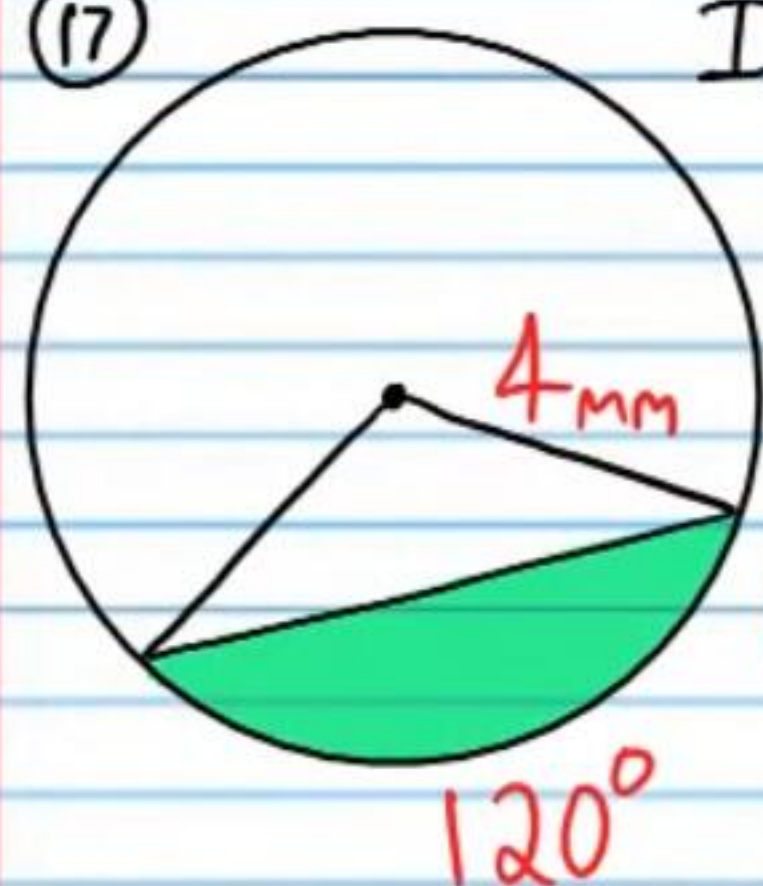
$$A_{\text{tri}} = 2$$

$$III) A_{\text{seg}} = A_{\text{sector}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 3.1415 - 2$$

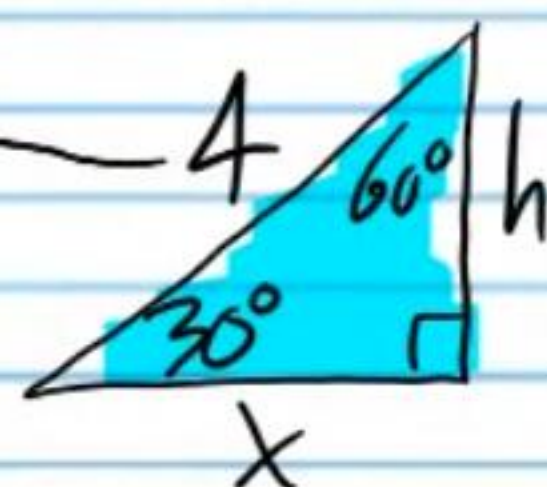
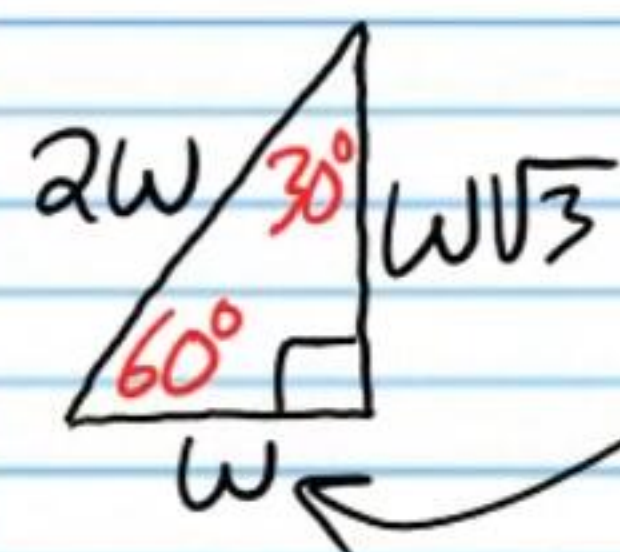
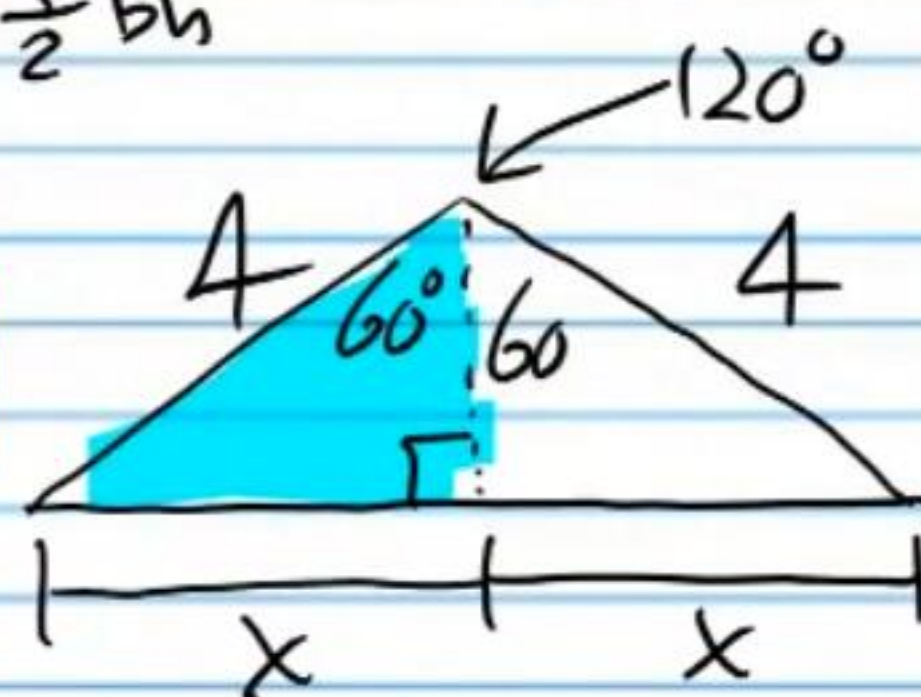
$$A_{\text{seg}} \approx 1.14 \text{ m}^2$$

⑪ I) $A_{\text{sector}} = \frac{120}{360} \pi (4)^2 \approx 16.7551$



II) $A_{\text{triangle}} = \frac{1}{2}bh$

Height
 $h = 2$



Base
 $x = 2\sqrt{3}$
 $b = 2x$
 $b = 2(2\sqrt{3})$
 $b = 4\sqrt{3}$

$A_{\text{tri}} = \frac{1}{2}(4\sqrt{3})(2)$

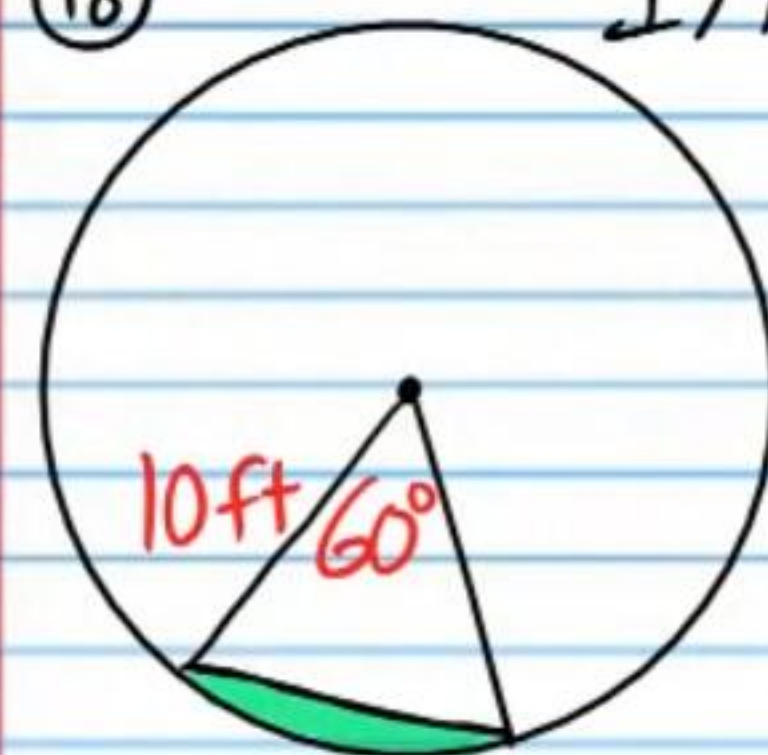
$A_{\text{tri}} \approx 6.9282$

III) $A_{\text{seg}} = A_{\text{sector}} - A_{\text{tri}}$

$A_{\text{seg}} \approx 16.7551 - 6.9282$

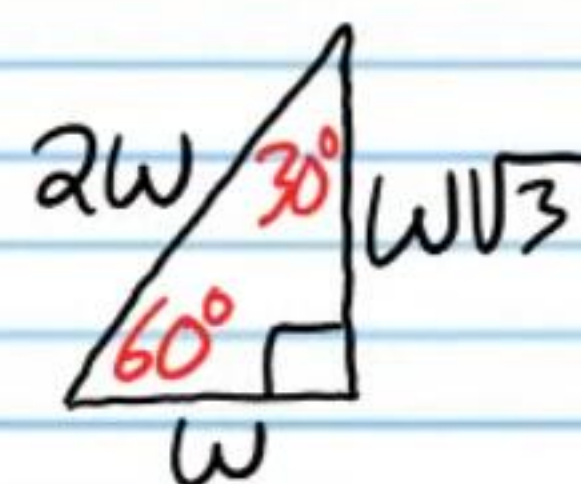
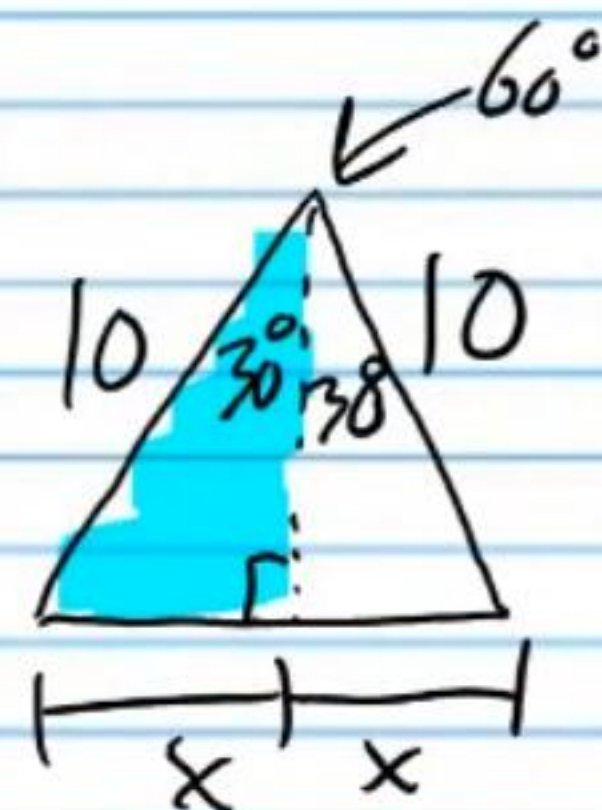
$A_{\text{seg}} \approx 9.83 \text{ mm}^2$

⑫ I) $A_{\text{sector}} = \frac{60}{360} \pi (10)^2 \approx 52.3598$



II) $A_{\text{triangle}} = \frac{1}{2}bh$

Height
 $h = 5\sqrt{3}$



Base
 $x = 5$
 $b = 2x$
 $b = 2(5)$
 $b = 10$

$A_{\text{tri}} = \frac{1}{2}(10)(5\sqrt{3})$

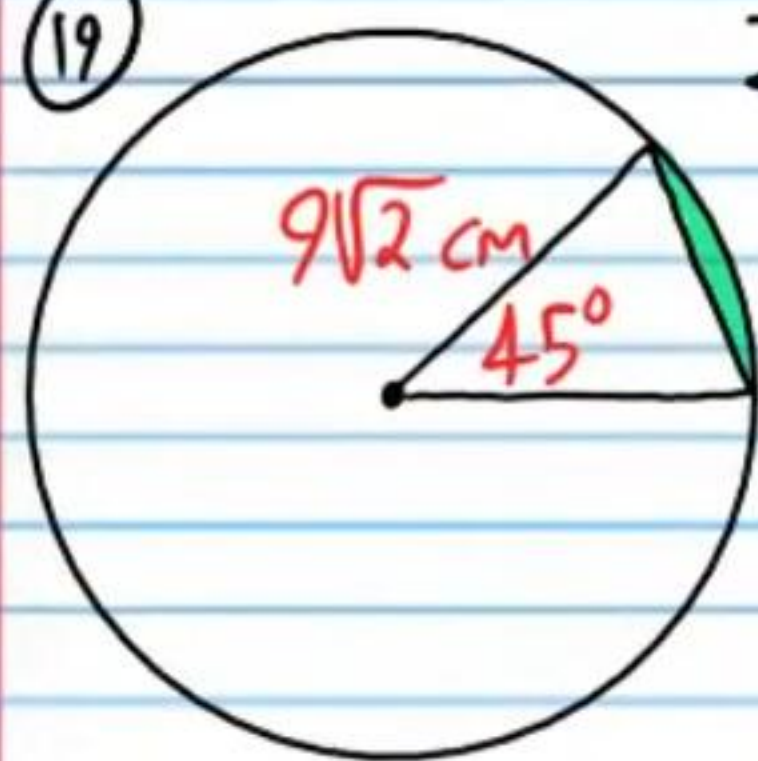
$A_{\text{tri}} \approx 43.3012$

III) $A_{\text{seg}} = A_{\text{sector}} - A_{\text{tri}}$

$A_{\text{seg}} \approx 52.3598 - 43.3012$

$A_{\text{seg}} \approx 9.06 \text{ ft}^2$

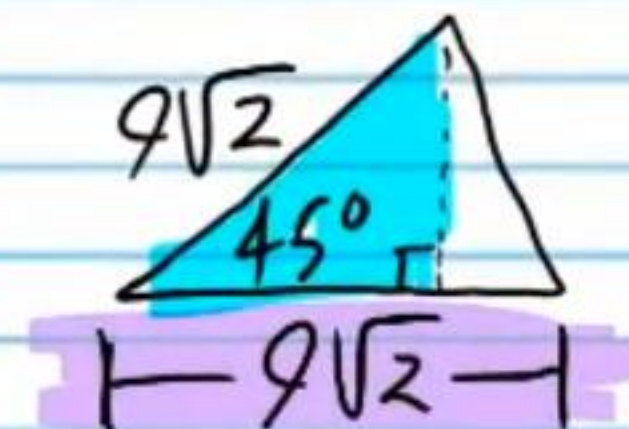
(19)



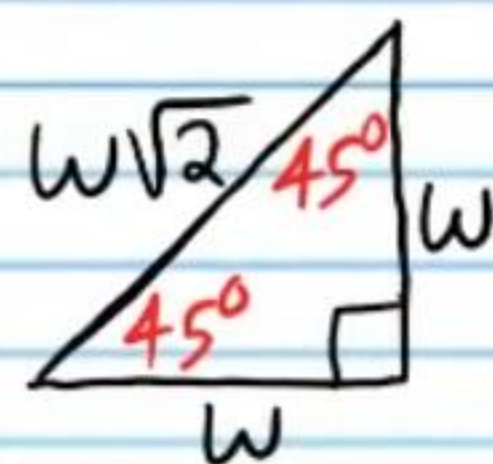
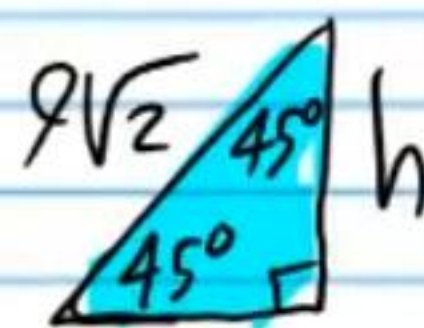
$$I) A_{\text{sect}} = \frac{45}{360} \pi (9\sqrt{2})^2 \approx 63.6172$$

$$II) A_{\text{tri}} = \frac{1}{2}bh$$

Height
 $h=9$



Base
 $b=9\sqrt{2}$



$$A_{\text{tri}} = \frac{1}{2}(9\sqrt{2})(9)$$

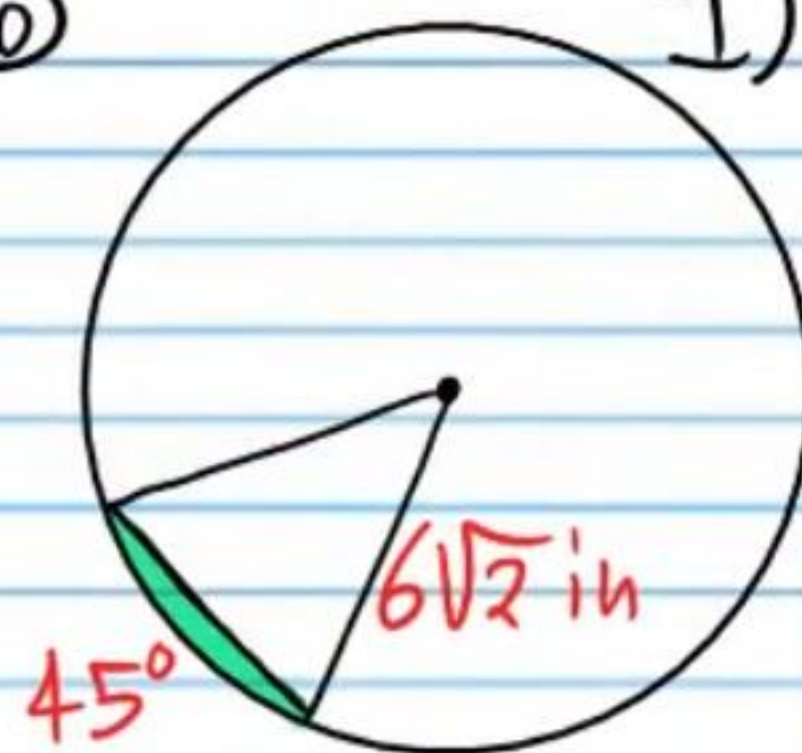
$$A_{\text{tri}} \approx 57.2756$$

$$III) A_{\text{seg}} = A_{\text{sect}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 63.6172 - 57.2756$$

$$A_{\text{seg}} \approx 6.34 \text{ cm}^2$$

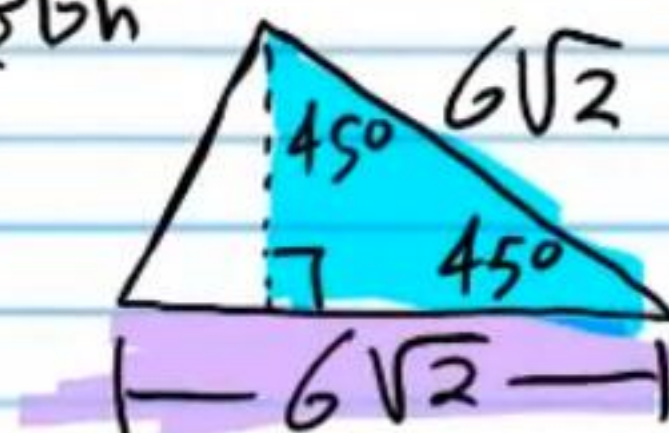
(20)



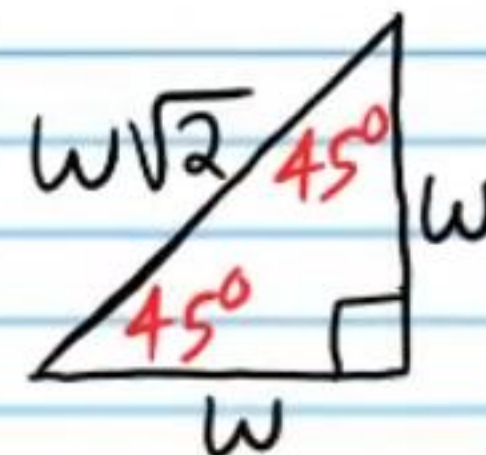
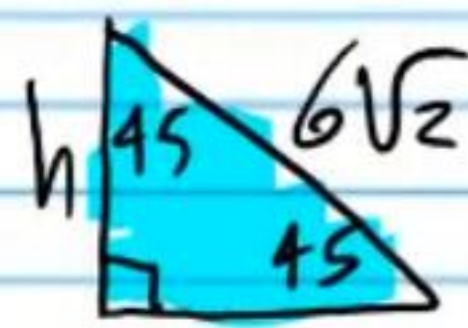
$$I) A_{\text{sector}} = \frac{45}{360} \pi (6\sqrt{2})^2 \approx 28.2743$$

$$II) A_{\text{triangle}} = \frac{1}{2}bh$$

Height
 $h=6$



Base
 $b=6\sqrt{2}$



$$A_{\text{tri}} = \frac{1}{2}(6\sqrt{2})(6)$$

$$A_{\text{tri}} \approx 25.4558$$

$$III) A_{\text{seg}} = A_{\text{sect}} - A_{\text{tri}}$$

$$A_{\text{seg}} \approx 28.2743 - 25.4558$$

$$A_{\text{seg}} \approx 2.82 \text{ in}^2$$